Worksheet 1

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1. Aim/Overview of the practical:

A. To understand the usage of fundamental Python libraries for Machine Learning
B. To perform comprehensive data preprocessing on the Titanic dataset.

2. Apparatus :

- Computer/Laptop
- Python 3.x
- Jupyter Notebook or Google Colab
- Titanic Dataset (CSV)
- Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

3. Code:

Part A: Implementation of Python Basic Libraries

1. Math Library

```
import math

print("Square Root:", math.sqrt(9))

print("factorial", math.factorial(4))

print("Power", math.pow(4,2))

print("ceil", math.ceil(5.66))

print("Floor", math.floor(5.66))

print("Pi value", math.pi)

print("Euler's number", math.e)
```

```
Square Root: 3.0
factorial 24
Power 16.0
ceil 6
Floor 5
Pi value 3.141592653589793
Euler's number 2.718281828459045
```

2. NumPy Library

```
import numpy as np

arr = np.array([2,4,6,8,10])

print("Array",arr)

print("mean:",np.mean(arr))

print("Sum:",np.sum(arr))

print("Max:",np.max(arr))

matrix = np.array([[1,2],[3,5]])

print("Matrix:\n",matrix)

print("transpose:\n",matrix.T)
```

```
Array [ 2  4  6  8 10]
mean: 6.0
Sum: 30
Max: 10
Matrix:
 [[1 2]
 [3 5]]
transpose:
 [[1 3]
 [2 5]]
```

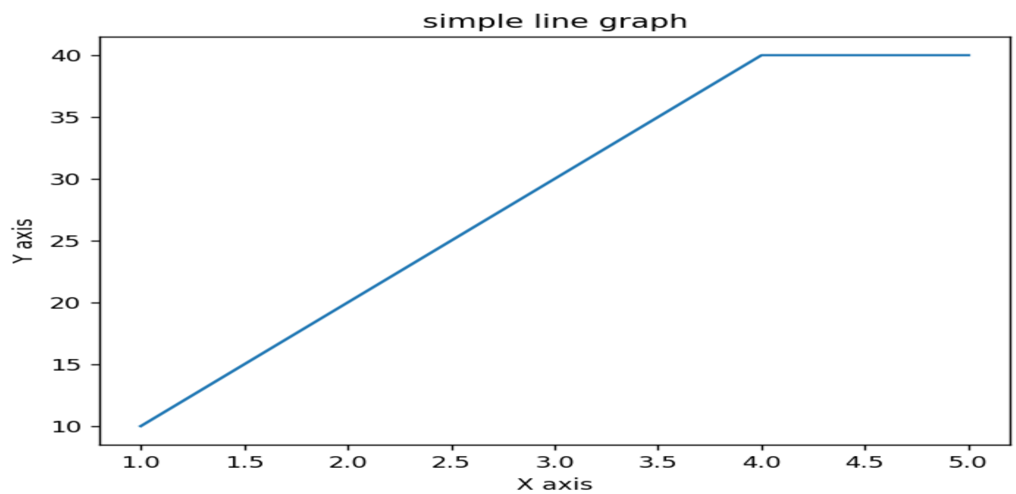
3. Matplotlib Library

```
import matplotlib.pyplot as plt

x =[1,2,3,4,5]

y =[10,20,30,40,40]

plt.plot(x,y)

plt.xlabel("X axis")
```

plt.ylabel("Y axis")

plt.title("simple line graph")

plt.show()



4. SciPy Library

from scipy import stats

data =[5,2,4,7,9]

print("Mean;",stats.tmean(data))

print("Median;",stats.scoreatpercentile(data,50))

print("Mode;",stats.mode(data))

```
Mean; 4.833333333333333
Median; 4.5
Mode; ModeResult(mode=np.int64(2), count=np.int64(2))
```

5. Seaborn Library

import seaborn as sns

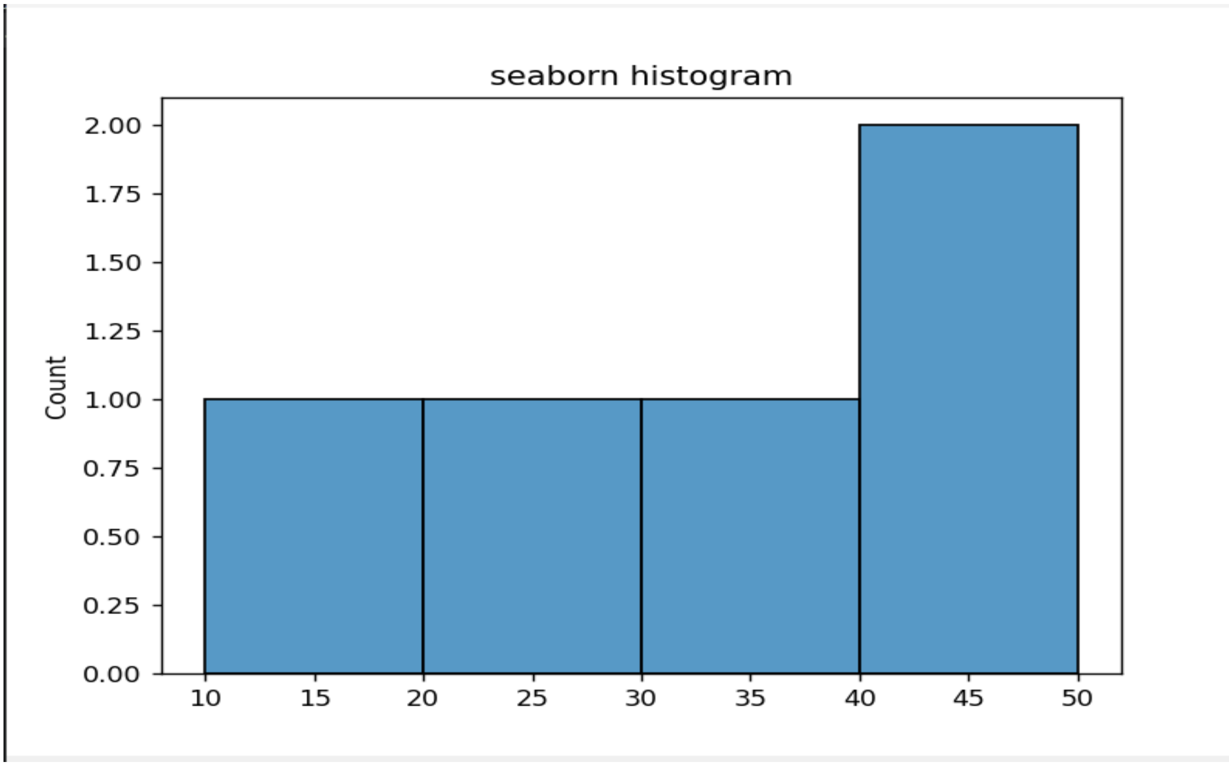
import matplotlib.pyplot as plt

data =[10,20,30,40,50]

sns.histplot(data)

plt.title("seaborn histogram")

plt.show()



Part B: Data Preprocessing on Titanic Dataset

Step 1: Import the required libraries

import numpy as np

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder

from sklearn.preprocessing import StandardScaler

Step 2: Load the Titanic dataset

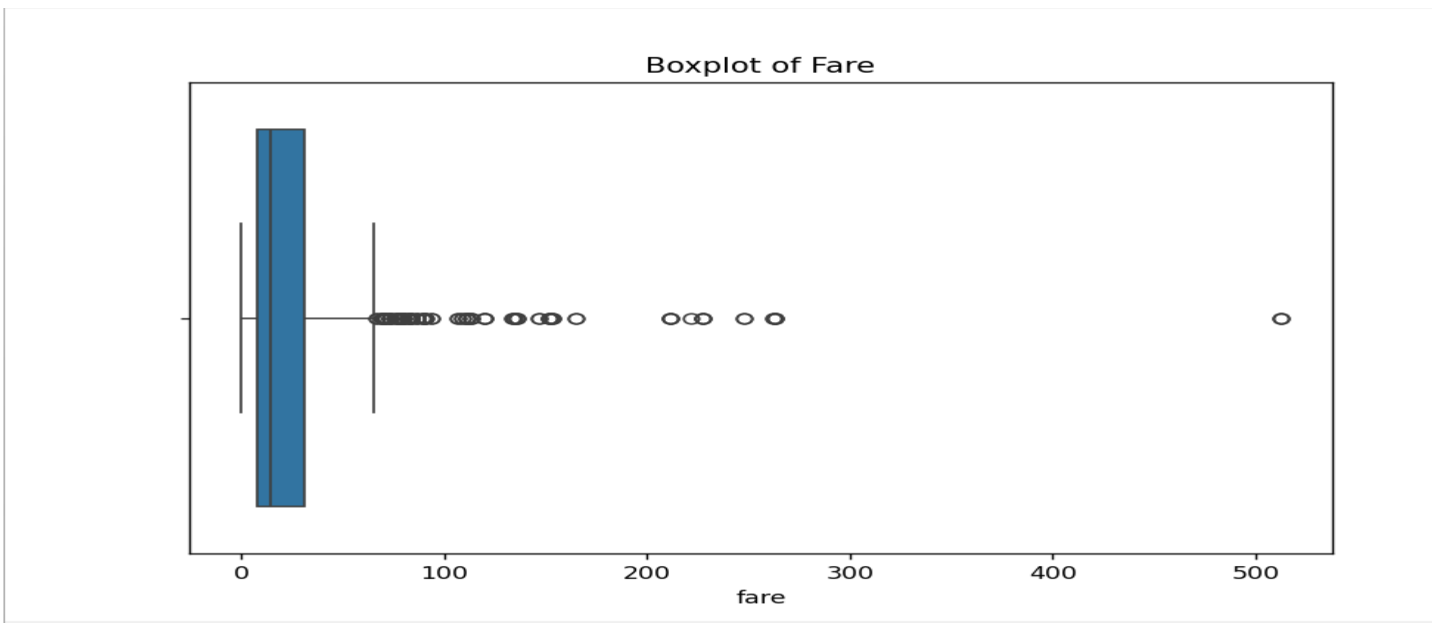
df = sns.load_dataset('titanic')

```
# Display the first few rows of the dataset
print("First Five Rows:")
print(df.head())
# Count the number of missing values in each column
print("\nMissing Values:")
print(df.isnull().sum())
# Replace missing values in the 'age' column with the median
df['age'] = df['age'].fillna(df['age'].median())
# Replace missing values in the 'embarked' column with the mode
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
# Remove the 'deck' column due to many missing values
df.drop(columns=['deck'], inplace=True)
# Remove any remaining missing values
df.dropna(inplace=True)
# Display boxplot to identify outliers in the 'fare' column
plt.figure(figsize=(8, 5))
sns.boxplot(x=df['fare'])
plt.title("Boxplot of Fare")
plt.show()
# Calculate first and third quartiles
Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)
# Calculate interquartile range
IQR = Q3 - Q1
```

```
# Define lower and upper limits
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
# Remove outliers from the 'fare' column
df = df[(df['fare'] >= lower) & (df['fare'] <= upper)]
# Encode categorical columns
encoder = LabelEncoder()
categorical_columns = ['sex', 'embarked', 'class', 'who', 'embark_town', 'alive']
for col in categorical_columns:
    if col in df.columns:
        df[col] = encoder.fit_transform(df[col])
# Create StandardScaler object
scaler = StandardScaler()
# Scale the 'age' and 'fare' columns
df[['age', 'fare']] = scaler.fit_transform(df[['age', 'fare']])
# Remove duplicate records
df.drop_duplicates(inplace=True)
# Display dataset information
print("\nDataset Information:")
print(df.info())
# Display first five records
print("\nProcessed Dataset:")
print(df.head())
```

OUTPUT:

```
First Five Rows:
   survived  pclass     sex   age  sibsp  parch     fare embarked  class    who  adult_male deck  embark_town alive  alone
0         0       3    male  22.0      1      0   7.2500        S  Third    man        True  NaN  Southampton    no  False
1         1       1  female  38.0      1      0  71.2833        C  First  woman       False    C    Cherbourg   yes  False
2         1       3  female  26.0      0      0   7.9250        S  Third  woman       False  NaN  Southampton   yes   True
3         1       1  female  35.0      1      0  53.1000        S  First  woman       False    C  Southampton   yes  False
4         0       3    male  35.0      0      0   8.0500        S  Third    man        True  NaN  Southampton    no   True

Missing Values:
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```



```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
Index: 665 entries, 0 to 890
Data columns (total 14 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     665 non-null    int64
 1   pclass       665 non-null    int64
 2   sex          665 non-null    object
 3   age          665 non-null    float64
 4   sibsp        665 non-null    int64
 5   parch        665 non-null    int64
 6   fare         665 non-null    float64
 7   embarked     665 non-null    object
 8   class        665 non-null    category
 9   who          665 non-null    object
 10  adult_male   665 non-null    bool
 11  embark_town  665 non-null    object
 12  alive        665 non-null    object
 13  alone        665 non-null    bool
dtypes: bool(2), category(1), float64(2), int64(4), object(5)
memory usage: 64.4+ KB
None

Processed Dataset:
   survived  pclass     sex       age  sibsp  parch      fare embarked  class    who  adult_male  embark_town alive  alone
0         0       3    male -0.528321      1      0 -0.779117        S  Third    man        True  Southampton    no  False
2         1       3  female -0.215182      0      0 -0.729373        S  Third  woman       False  Southampton   yes   True
3         1       1  female  0.489381      1      0  2.599828        S  First  woman       False  Southampton   yes  False
4         0       3    male  0.489381      0      0 -0.720161        S  Third    man        True  Southampton    no   True
5         0       3    male -0.058613      0      0 -0.690071        Q  Third    man        True   Queenstown    no   True
```

5. Learning outcomes (What I have learnt):

- Learned to use Python libraries for data analysis.
- Learned to clean and prepare the Titanic dataset.
- Learned to visualize data using simple graphs and charts.